

# TrES-5b

R-0930

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_170630 · created 2026-07-29T11:05:07+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457934.9057 +/- 0.0024 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.144 +/- 0.04                  |
| Transit depth (Rp/Rs)^2                           | 2.09 +/- 1.16 [%]               |
| Orbital Inclination (inc)                         | 82.48 +/- 2.77                  |
| Airmass coefficient 1 (a1)                        | 1.277 +/- 0.02                  |
| Airmass coefficient 2 (a2)                        | -0.21 +/- 0.1                   |
| Scatter in the residuals of the lightcurve fit is | 1.56 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 2.5                             |
| Optimal Annulus                                   | 8.09                            |
| Transit Duration (day)                            | 0.054 +/- 0.025                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                         |
|---------------------------|-----------------------------------------|
| Rp/R■ fit vs published    | 0.144 ± 0.04 vs 0.142 (deviation 0.04σ) |
| Duration fit vs published | 0.054 d vs 0.0758 d (deviation 0.78σ)   |
| Mid-time O–C              | -5.9 min                                |
| Depth SNR                 | 3.2                                     |
| Residual scatter          | 1.56 %                                  |

## Light curve

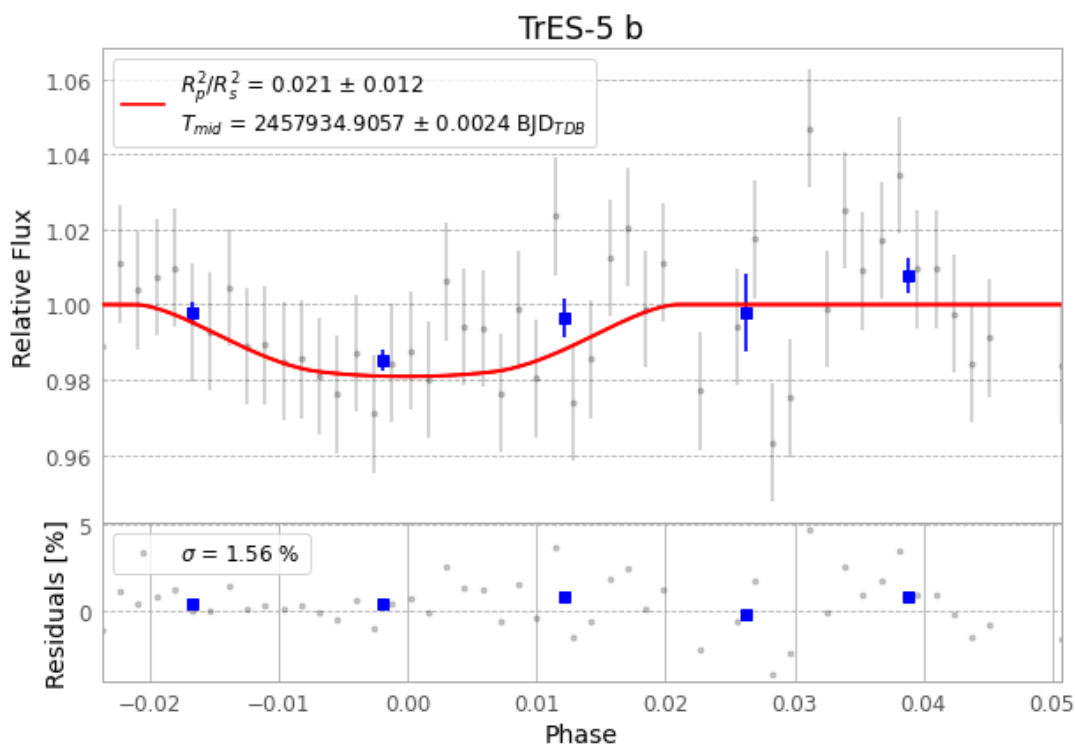


Figure 1 — FinalLightCurve\_TrES-5 b\_2017-06-30.png (SHA-256 155c4dda2da319b9...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [226, 278], 1 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 646e7638a56b1bcd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

56 FITS frames (37 MB), TRES-5170630085112.FITS → TRES-5170630115413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-170630.log (396c32c4777b...), FinalLightCurve\_TrES-5 b\_2017-06-30.png (155c4dda2da3...), FinalLightCurve\_TrES-5 b\_2017-06-30.csv (26b865e610d4...)

## Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 11:05Z.